



X-RAY WITH A RESTRICT MASK — THE WRECK

Predict what's behind the machine — and the honest stress test of LaRI's weak domain.

FILE `atlas_xray_wreck_workflow.json` VERIFIED 2026-07-16

THE STORY

What's behind the rusty machine? AtlasPredictHiddenGeometry (LaRI) predicts the surface stack each camera ray pierces and substitutes the occluder's pixels with the first layer that clears it. Unconstrained, it wanted to substitute 50.6% of the frame — including plainly visible background. The fix is one wire: a SAM3 segment of the machine itself into `restrict_mask`. Substitution drops to 23.8% — the machine only, everything else untouched.

This plate is deliberately the honest one: an isolated object on open moor is LaRI's documented WEAK domain (architecture is its strength — see the NYC X-ray variant, which scored the set's best registration). The node's report states measured coverage every run, so you know how much to trust before a single frame renders. LaMa provides the reveal pixels; AtlasRenderFix ships bypassed (it needs a Docker environment) but slots between the baked frames and the MP4 when built.

IN THE VIEWPORT — PROJECTED VS. THE GREY MESH



SOURCE PLATE

atlas_00025.png · coastal wreck · 7680×4512 — the single still photograph everything below is reconstructed from.



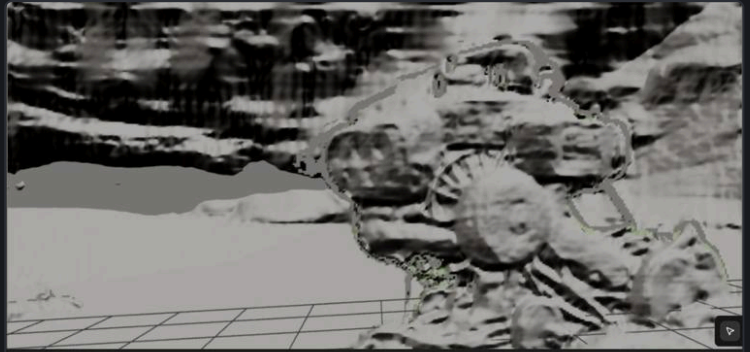
PROJECT ON

The photo reassembled on the recovered camera — texels assigned by ray, perfect from this view.



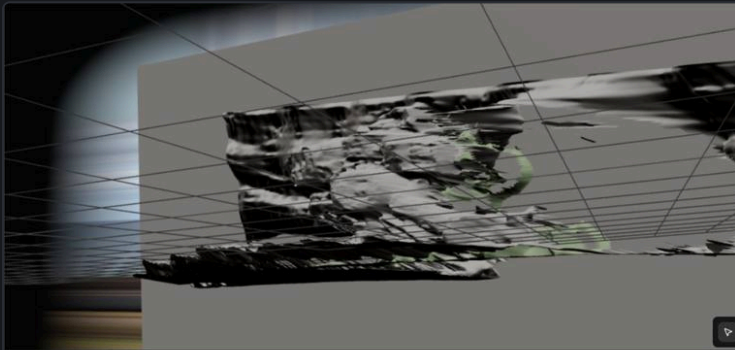
X-RAY PROVENANCE

Red tint = surface whose depth was INVENTED by the hidden-geometry prediction — exactly what you're trusting the model for, nothing more.



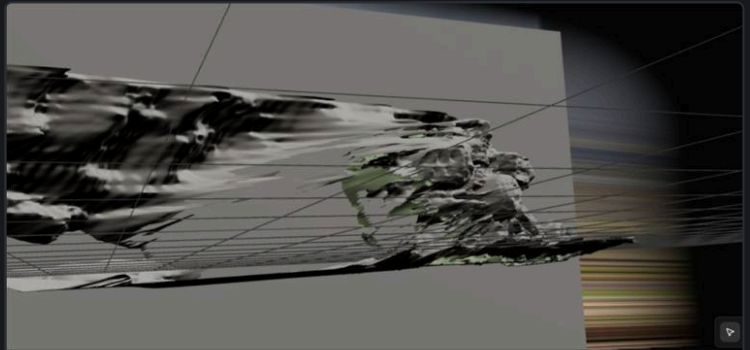
PROJECT OFF

The same geometry as a grey mesh, recovered view — what the depth-derived surfaces actually are.



GREY · ORBIT LEFT

Orbited off-axis: the reconstructed cone. Coverage ends where the camera never saw.



GREY · ORBIT RIGHT

The opposite angle — silhouette tears read as deliberate holes, not errors.

NODES ON STAGE

AtlasPredictHiddenGeometry

SAM3Segment

AtlasDepthMap

AtlasCleanPlateLayer

AtlasRenderFix (bypassed)

LaMa inpaint

RETUNE ON YOUR PLATE

- ▶ **The SAM3 prompt IS the workflow.** 'rusty machine' → describe YOUR occluder. The restrict mask is not optional in practice — the unrestricted number above is why.
- ▶ **Expect partial coverage on open terrain.** Read the node's report (registration rel-MAD, coverage, layer histogram) before judging the result.
- ▶ **AtlasRenderFix stays bypassed** until you've built the fixer-spike-env Docker image and cloned nv-tlabs/Fixer. Frames pass straight through otherwise.

RUN-VERIFIED

RUN ✓ Restrict mask cut substitution from 50.6% → 23.8% of frame; registration rel-MAD 0.190 ('good'). The moor background's partial coverage is the documented, expected result.

REQUIREMENTS

ATLAS_EXPERIMENTAL=1 · LaRI clone · ComfyUI-RMBG · LaMa · Docker only for RenderFix

Atlas Camera v0.6.0 beta · [examples/showcase](#)

[Master field guide](#) →



ATLAS CAMERA · RECOVER THE CAMERA FROM A SINGLE STILL